

CTU-Chat: Supervisor-Routed Multi-Agent RAG with Heterogeneous Knowledge Allocation for University Counseling

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Abstract. Student counseling in higher education draws on heterogeneous institutional knowledge, including relational curricula, cohort-specific tuition schedules, statutory exemption criteria, scholarship policies, and narrative administrative regulations. This setting poses a challenge for uniform retrieval and reasoning pipelines, as different forms of institutional knowledge may benefit from different representations, retrieval mechanisms, and processing strategies. We present CTU-Chat, a centralized supervisor-routed, domain-specialized multi-agent system for Vietnamese university counseling. The supervisor directs each query to one of four specialists, while each specialist is assigned an appropriate knowledge source, retrieval path, and set of tools. Relational academic information is handled through a knowledge graph; structured tuition and scholarship information is supported by dedicated lookup and calculation functions; and narrative regulations are retrieved through hybrid document search. Evaluation shows that CTU-Chat achieves 97.0% supervisor routing accuracy, while its proposed retrieval configuration reaches 0.7500 Hit@1 and 0.8354 MRR@10 on a redesigned 100-query held-out test set. These findings suggest that centralized coordination with domain-specific knowledge allocation is a promising approach for heterogeneous university counseling.

Keywords: Multi-Agent Systems · Domain-Specialized Agents · Centralized Supervisor Routing · Heterogeneous Knowledge Allocation · Retrieval-Augmented Generation · University Counseling

1 Introduction

Can Tho University (CTU) is a key higher education institution in the Mekong Delta, with nearly 49,000 undergraduate students in the 2025–2026 academic year [1]. At an institution of this scale, advisory needs span a wide range of academic and administrative areas. Relevant regulations and policies may also change

over time. Prior research on automated advisory systems includes CAAS, which supports chat- and voice-based admission counseling for information-technology programs [10], and REBot, which focuses on academic-regulation advising through dense retrieval and category-guided graph reasoning [9]. Their reported evaluations illustrate the value of task-specific retrieval and processing strategies, but remain centered on individual advisory domains. This motivates CTU-Chat as a coordinated multi-domain architecture for heterogeneous university counseling.

University counseling presents challenges not only because it covers multiple domains, but also because institutional knowledge is represented in different forms. (1) **Academic curricula** contain hierarchical structures and prerequisite relationships; (2) **tuition schedules** are organized by admission cohort and field of study; (3) **tuition exemption policies** and (4) **scholarship policies** combine eligibility conditions with specific support rates and thresholds; while (5) **general administrative regulations** are presented in lengthy narrative documents. Therefore, no single representation or retrieval method is suitable for every domain, motivating representation-specific evidence acquisition in CTU-Chat.

RAG and GraphRAG augment LLMs with external documents and knowledge graphs [8,5,3], while agentic architectures support tool invocation and multi-agent coordination [20,18,16]. However, these approaches do not by themselves determine how heterogeneous institutional evidence should be represented, routed, and processed. Moreover, known limitations in language-model arithmetic [14] motivate deterministic calculation for rule-based financial queries.

These challenges highlights three open research gaps in automated university counseling: (1) limited domain-specialized multi-agent orchestration; (2) uniform representation of heterogeneous institutional knowledge; and (3) the lack of representation-aware evidence routing. These gaps are examined in greater detail in Section 2.

The objective of this study is to develop and evaluate CTU-Chat for heterogeneous university counseling. To address the identified gaps, we formulate three research questions:

- **RQ1 (Domain-Specialized Multi-Agent Architecture):** How effectively can a centralized supervisor coordinate domain-specialized agents with isolated knowledge spaces and constrained tool access for heterogeneous university counseling tasks?
- **RQ2 (Heterogeneous Knowledge Allocation & Modular Ablations):** What differences in context and answer quality are observed between the full heterogeneous configuration, uniform retrieval baselines, and its component ablations?
- **RQ3 (Representation-Aware Retrieval Effectiveness):** How does the complete representation-aware evidence configuration compare with sparse, dense, hybrid, and reranked baselines in retrieval effectiveness, both overall and across counseling domains?

We hypothesize that reliable university counseling is better supported by a supervisor-routed multi-agent architecture in which each specialist is restricted

to domain-appropriate knowledge representations, retrieval paths, and tools. To investigate this hypothesis and the research questions above, we propose CTU-Chat, which coordinates four domain-specialized agents for academic, financial, scholarship, and general administrative inquiries within a unified framework combining graph-based retrieval, structured data access, deterministic calculation, and hybrid document retrieval.

The remainder of this paper is organized as follows. **Section 2** reviews related work and summarizes the research gaps. **Section 3** presents the CTU-Chat architecture, evidence-acquisition methods, and domain-specialized agents. **Section 4** reports the experimental setup, results, and discussion. Finally, **Section 5** concludes the paper and outlines future work.

2 Related Work

2.1 Vietnamese NLP Foundations and Linguistic Specifics

Unlike whitespace-delimited languages, Vietnamese is an isolating, tonal language where white spaces separate syllables rather than words [11], with over 85% of lexical units forming multi-syllabic compound words (e.g., “*học phí*” [tuition fee]). While domain-adapted bi-encoders like PhoBERT capture semantic paraphrasing, Vietnamese higher education decrees rely heavily on rigid Sino-Vietnamese legal terminology and alphanumeric cohort codes. This dual characteristic motivates coupling syllable-aware dense embeddings with exact lexical token matching to mitigate both semantic drift and alphanumeric recall failures.

2.2 Sparse and Dense Retrieval

Lexical Retrieval (BM25): Sparse keyword search using BM25 [15] scores documents via term frequency (tf) and inverse document frequency (IDF) over an inverted index ($k_1 = 1.5, b = 0.75$). BM25 provides reliable exact token matching for rigid alphanumeric identifiers (decision codes, cohort labels, course codes), but cannot capture semantic paraphrases.

Dense Semantic Retrieval: Dense retrieval systems [7,8] map queries and passages into embedding spaces using bi-encoders, computing relevance via vector similarity. While dense models capture paraphrastic intent, rare alphanumeric identifiers and out-of-vocabulary regulatory tokens can remain difficult and may introduce cross-cohort noise, motivating hybrid retrieval.

2.3 Hybrid Retrieval and Reciprocal Rank Fusion

To merge BM25 and dense candidates without score calibration, Reciprocal Rank Fusion (RRF) [2] sorts documents by reciprocal position: $s_{\text{RRF}}(d) = \sum_{s \in \mathcal{M}} \frac{1}{k + r_s(d)}$ ($k = 60$), while neural cross-encoders [12,19] jointly encode query–passage pairs for reranking. Adaptive-RAG further demonstrates that retrieval strategies can be selected according to question complexity [6]. These approaches motivate dynamic retrieval, but do not directly allocate institution-specific relational, tabular, and narrative knowledge to isolated specialist paths.

2.4 Knowledge Graphs and GraphRAG

Knowledge graphs represent domain knowledge as structured triples (h, r, t) [13]. GraphRAG [3] leverages graph topology to uncover multi-hop associative paths escaping flat chunking, benefiting curriculum structures and prerequisite chains. However, universal graph extraction may introduce spurious or hallucinated relationships, especially when narrative documents lack explicit relational structure. Therefore, this work adopts selective rather than universal graph construction.

2.5 Agentic RAG and Multi-Agent Orchestration

ReAct demonstrates interleaved reasoning and action, AutoGen provides a framework for multi-agent conversation, and Toolformer studies when and how language models invoke external APIs [20,18,16]. For institutional counseling, these capabilities raise design risks when applied without explicit constraints: broad tool exposure can increase tool and parameter errors; peer-to-peer exchanges can increase coordination cost; a shared index can mix evidence across domains; and a common retrieval path can obscure opportunities for direct graph traversal or structured lookup. These risks motivate the bounded, supervisor-routed specialization evaluated in this work.

2.6 Educational and University Counseling Systems

Educational chatbots assist in advising and learning support [17], with recent systems like REBot [9] combining text retrieval with category-guided GraphRAG for academic regulations. CTU-CHAT departs from prior advisory systems across three key aspects: (1) *Task Scope*: while representative prior advisory systems often emphasize single-domain text-QA or focus primarily on narrative regulations, CTU-CHAT models 113 curricula as relational property graphs and incorporates deterministic arithmetic engines (`tin_h_toan_hoc_phi`, `tin_h_tien_hoc_bong`) for cohort-dependent tuition, exemptions, and scholarship thresholds; (2) *Multi-Agent Routing*: rather than a monolithic classifier-guided pipeline, a LangGraph supervisor dynamically routes queries to four bounded specialists with isolated evidence spaces and asymmetric tool privileges; and (3) *Metric Incommensurability*: prior benchmarks evaluate traditional classification accuracy and binary Q&A F_1 -scores, precluding direct numerical comparison with CTU-CHAT, which evaluates end-to-end generative fidelity via Ragas and bounded multi-agent tool execution benchmarks.

2.7 Comparative Synthesis and Research Gaps

Synthesizing established retrieval and reasoning paradigms highlights clear operational trade-offs for university counseling [2,3,6,7,15,16,18,20]. While individual techniques resolve isolated needs—lexical matching for exact regulatory tokens, dense embeddings for semantic paraphrasing, and GraphRAG for relational

prerequisites—none simultaneously address the tripartite challenge of multi-domain institutional advising.

This analysis exposes three operational gaps in automated university counseling: (1) *Lack of bounded multi-agent orchestration*: monolithic pipelines and unconstrained agent meshes lack execution guardrails, frequently triggering tool mis-selection and parameter hallucination; (2) *Limitations of uniform representation*: forcing relational curricula, tabular tuition matrices, and narrative legal text into a single retrieval space can cause prerequisite fragmentation and semantic drift; and (3) *Absence of representation-aware routing*: dynamic steering mechanisms that allocate queries across isolated representation pathways while offloading statutory arithmetic to deterministic engines remain largely unaddressed in higher-education administrative counseling. These structural requirements directly motivate the centralized supervisor-routed architecture detailed in Section 3.

3 Proposed Model

This section presents the architecture, algorithms, and evidence-acquisition mechanisms of CTU-CHAT. The proposed model is organized around three design principles: *centralized agent coordination*, in which a supervisor dispatches each query to one domain agent without peer-to-peer delegation; *heterogeneous knowledge allocation*, which assigns each information structure to a corresponding evidence representation; and *representation-aware evidence routing*, which selects either a direct graph-tool path or an intent-scoped retrieval path according to the assigned domain. Here, each *specialist* represents a bounded domain agent configured with dedicated prompt constraints, an assigned evidence store, and a restricted tool registry.

3.1 Overall Proposed Architecture

Figure 1 illustrates the system architecture of CTU-CHAT, structured across three coordinated tiers: (1) the *Interaction and Centralized Supervisor Tier*, (2) the *Domain-Specialized Multi-Agent Core Layer*, and (3) the *Heterogeneous Knowledge Storage and Tool Tier*.

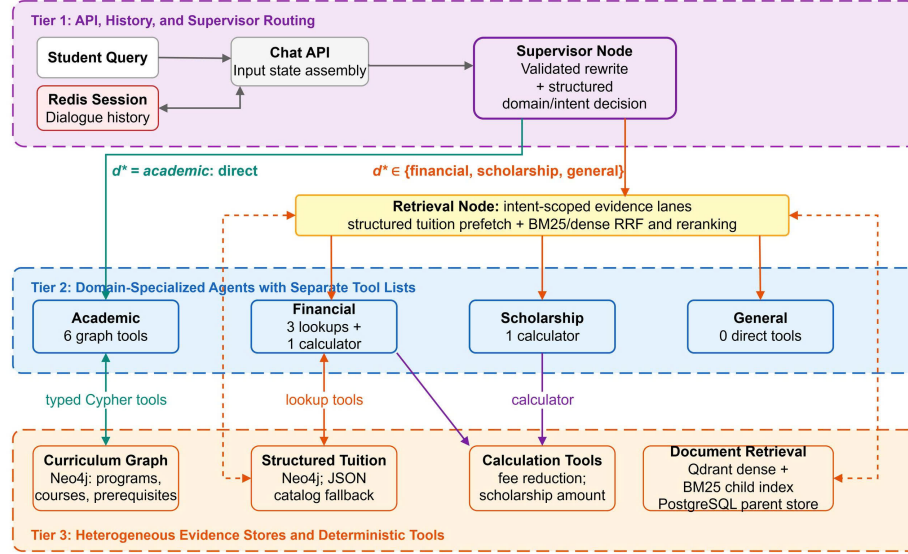


Fig. 1. Implemented CTU-CHAT architecture. The Chat API loads dialogue history from Redis and supplies it in the graph input state. The first conditional edge sends academic queries directly to the graph-enabled agent and sends the other domains through intent-scoped retrieval; the second edge dispatches the retrieved evidence to the selected domain-specialized agent. Solid agent–resource links denote direct tool access, whereas dashed links denote evidence acquisition performed by the retrieval node.

Application-Level Specialization of LangGraph StateGraph: Rather than modifying the underlying LangGraph engine, CTU-CHAT establishes an application-level orchestration policy governed by a typed state contract, isolated execution paths, and partitioned tool registries. Following supervisor dispatch, the four domain specialists operate independently without peer-to-peer delegation, selecting actions strictly from their construction-time tool bindings.

- **Node-scoped state updates:** `AgentState` defines explicit write boundaries: the supervisor writes to the routing fields (`search_query`, `next_agent`, and `routing_decision`); the retrieval node updates `context`; and each specialist returns `response`. This contract enforces transparent data ownership and mitigates cross-domain context pollution.
- **Two-Tier Conditional Routing Edges:** Instead of letting agents dynamically negotiate execution flow, the orchestration topology is governed by deterministic conditional functions:

$$\text{Edge}_{\text{sup}}(\mathcal{S}) = \begin{cases} \text{'academic_agent'}, & \text{if } \mathcal{S}[\text{'next_agent'}] = \text{'academic'} \\ \text{'retrieval'}, & \text{otherwise,} \end{cases} \quad (1)$$

bypassing document retrieval for queries assigned to the academic agent. At the retrieval exit, $\text{Edge}_{\text{ret}}(\mathcal{S})$ dispatches the retrieved context to the selected financial, scholarship, or general agent.

- **Acyclic outer orchestration:** The StateGraph has no peer-to-peer agent edges. Each selected domain-specialized agent returns to END, so the outer routing path cannot cycle. ReAct tool interactions inside an agent remain distinct from this outer graph topology.
- **Domain-specialized agents with asymmetric tools:** The academic, financial, and scholarship agents are constructed with separate prompts and tool lists, while the general agent has no direct tool (as detailed in Figure 1). This confines tool selection to the agent selected by the supervisor and keeps domain-specific evidence paths separate, reducing decision space pollution.

3.2 Neo4j Knowledge Graph Specification and Cypher Tool Mechanics

Schema defined by the ingestion layer: The Neo4j graph models relational dependencies across academic counseling using a simplified schema, comprising over 1,500 entity nodes and 3,800 relational edges across 113 undergraduate curricula and statutory fee schedules. It incorporates 10 node labels (7 for curriculum: Program, Course, Block, etc.; and 3 for tuition: TuitionFee, TuitionPolicy, ExemptionBasisRate) interconnected via 13 relationship types (HAS_BLOCK, CONTAINS, REQUIRES, PARALLEL_WITH, HAS_PLO, HAS_PEO, REALIZED_BY, HAS_TRACK, BELONGS_TO_TRACK, HAS_JOB_POSITION, HAS_TUITION, GOVERNED_BY, and HAS_EXEMPTION_BASIS). Narrative regulations remain in the document store unless linked to structured entities.

Cypher tool mechanics: The system does not ask the LLM to generate arbitrary Cypher. After the supervisor selects the academic agent, its ReAct policy chooses one of six typed tools and supplies the tool arguments. The tool normalizes identifiers or names and calls a fixed, parameterized Cypher query in the graph service. The tools implement three query classes:

- *Attribute lookup and search:* `tra_cuu_nganh` and `tim_nganh` match program codes or normalized program, unit, and course names; `so_sanh_nganh` applies the same lookup before comparing two programs.
- *Structural traversal:* `xem_chuoi_tien_quyet` follows REQUIRES paths up to ten hops and returns direct PARALLEL_WITH courses. `mon_chung_giua_nganh` computes a course intersection across two program–block paths, and `tim_nganh_co_mon` traverses those paths in reverse.
- *Structured financial lookup:* parameterized queries match TuitionFee, TuitionPolicy, and ExemptionBasisRate properties and their program relationships. This graph lookup is separate from the BM25–dense hybrid document retrieval path used for narrative regulations.

This design binds agent actions directly to deterministic graph operations: the LLM only selects the appropriate tool and supplies its arguments, while the query topology remains fixed and parameterized within the graph service. As a result, the multi-agent workflow reliably incorporates graph evidence while mitigating the syntactic and semantic fragility typically associated with unconstrained Cypher generation.

3.3 Algorithmic Foundations: Routing and Deterministic Arithmetic

Algorithm 1 specifies the supervisor-routing and evidence-path dispatch procedure.

Algorithm 1: Supervisor Routing and Conditional Evidence-Path Dispatch

Input: Student query q , Dialogue history $\mathcal{H}$, Shared state $\mathcal{S}$
Output: State updates and next graph node

```

1  $q_{\text{standalone}} \leftarrow q$ ;
2  $q_{\text{prev}} \leftarrow \text{MostRecentUserQuery}(\mathcal{H})$ ;
3 if ShouldRewriteQuery( $q, q_{\text{prev}}$ ) then
4    $q_{\text{cand}} \leftarrow \text{FastLLM.Rewrite}(q, q_{\text{prev}})$ ;
5   if ValidateRewrite( $q, q_{\text{cand}}, q_{\text{prev}}$ ) then
6      $q_{\text{standalone}} \leftarrow q_{\text{cand}}$ ;
7  $\langle d^*, \iota^* \rangle \leftarrow \text{StructuredSupervisor}(q)$ ;
8 if structured classification fails then
9    $d^* \leftarrow \text{'general'}$ ;  $\iota^* \leftarrow \text{'other'}$ ;
10 Update  $\mathcal{S}[\text{search\_query}] \leftarrow q_{\text{standalone}}$ ,  $\mathcal{S}[\text{next\_agent}] \leftarrow d^*$ , and
     $\mathcal{S}[\text{routing\_decision}] \leftarrow \text{QueryRoutingDecision}(\iota^*)$ ;
11 if  $d^* = \text{'academic'}$  then
12    $v_{\text{next}} \leftarrow \text{'academic\_agent'}$ ;
13 else
14   Build retrieval lanes from  $\iota^*$  and retrieve evidence;
15    $v_{\text{next}} \leftarrow d^*_{\text{agent}}$ ;
16 return  $\langle \mathcal{S}, v_{\text{next}} \rangle$ ;

```

A candidate rewrite is adopted only after validation; structured output restricts classification to declared labels with **general/other** as fallback; and conditional edges route academic queries directly to the agent while other domains pass through retrieval.

Deterministic Tuition Reduction Arithmetic: For fee-reduction requests, CTU-CHAT augments the LLM with the external deterministic tool `tinh_toan_hoc_phi`. The LLM interprets the request and supplies three parameters—the actual per-credit tuition T , the per-credit exemption basis B , and the percentage entitlement p —but does not perform the arithmetic through free-form text generation. The external tool validates the percentage range and applies the fixed calculation

$$D = B \frac{p}{100}, \quad P = \max(0, T - D). \quad (2)$$

where D is the reduction per credit and P is the remaining amount per credit. Monetary inputs are expected in VND per credit. The tool enforces $0 \leq p \leq 100$ and clamps P at zero; values outside the percentage range return an error.

Rather than a new general-purpose algorithm, the contribution lies in composing validated rewriting, constrained routing, representation-specific dispatch, and deterministic arithmetic within the multi-agent workflow.

3.4 Heterogeneous Knowledge Allocation and Hybrid Retrieval

This layer implements *heterogeneous knowledge allocation*: rather than forcing all institutional knowledge into a uniform document index, CTU-CHAT preserves relational curriculum knowledge in Neo4j, accesses cohort-dependent tuition data through structured lookup, and retains narrative regulations and scholarship policies in document retrieval.

Table 1 makes this policy explicit by separating the physical evidence representation from the mechanism used to access it at runtime.

Table 1. Decision criteria mapping query characteristics to evidence representation mechanisms.

Query characteristic	Evidence path	Rationale
Program/course attributes	Neo4j property lookup	Exact or normalized match
Prerequisites/shared courses	Neo4j traversal	Stored graph relationships
Regulation identifiers	BM25 child index	Exact token match
Paraphrased regulations	Hybrid RRF + reranker	Lexical + semantic signals
Fee/scholarship calculation	Python tool	Fixed formula outside LLM

At runtime, the StateGraph enforces this allocation. Academic queries bypass the document-retrieval node and enter the graph-enabled academic agent directly. Financial, scholarship, and general queries enter the retrieval node, where the supervisor-assigned intent selects metadata-filtered retrieval lanes before the resulting context is passed only to the assigned domain agent.

Hybrid Document Retrieval Path. For narrative regulations and scholarship criteria, evidence is acquired through the following pipeline:

1. *Child-level candidate generation.* A BM25 index searches child chunks, while Qdrant performs dense retrieval over their 768-dimensional embeddings.
2. *Parent resolution.* Parent identifiers associated with the retrieved child chunks are used to load the corresponding parent sections from PostgreSQL.
3. *Reciprocal Rank Fusion.* The dense- and BM25-derived parent rankings are combined as

$$s_{\text{RRF}}(d) = \sum_{s \in \{\text{bm25}, \text{dense}\}} \frac{1}{k + r_s(d)}, \quad k = 60.$$

4. *Cross-encoder reranking.* The fused parent candidates are reranked using BAAI/bge-reranker-v2-m3, and up to six parent sections are supplied to the selected domain agent.

Parent sections use a 2,800-character target with 100-character overlap; child chunks use a 400-character target with 50-character overlap.

4 Experimental Results

This section presents the empirical evaluation of CTU-CHAT.

4.1 Experimental Benchmark and Protocols

To support empirical evaluation across representative counseling scenarios, our evaluation uses a corpus of 200 administrative queries constructed via a document-grounded semi-automated protocol over official Can Tho University (CTU) statutes. Candidate queries covering diverse conversational phrasing (e.g., student colloquialisms, prerequisite rules, and fee schedules) were generated from statutory articles and reviewed against the institutional corpus to verify ground-truth intents, gold passages, and factual answers. The corpus is partitioned into a **Development Benchmark** ($N = 100$ queries), used for offline index calibration, prompt engineering, and qualitative failure analysis, and a separately constructed **Held-Out Test Set** ($N = 100$ queries). The held-out set follows a predefined complexity distribution comprising 40 direct single-hop, 20 within-domain multi-hop, 20 cross-domain, 10 entity-comparison, 5 temporal, and 5 adversarial queries. It contains 51 formal and 49 colloquial formulations; 56 queries have one gold source and 44 require multiple gold sources.

The construction audit used entity inventories separated from those in the development benchmark. It found no exact duplicate queries across the two splits and recorded a maximum cross-split 5-gram Jaccard similarity of 0.1875. Gold sources and required facts were reviewed against 244 institutional documents and 559 structured tuition records. The four single-domain subsets contain 19 Academic, 28 Financial, 17 Scholarship, and 16 General queries; the remaining 20 queries combine evidence across domains and are therefore analyzed separately rather than assigned to a single specialist domain. All primary metrics in Tables 2 and 4 are evaluated on this held-out set.

Execution and Evaluation Invariants: Generative and automated evaluation tasks were executed using Google Gemini 2.5 Flash Lite via Google Cloud Vertex AI under zero-temperature greedy decoding ($T = 0.0$). To account for subtle API-level non-determinism across cloud endpoints, each query was evaluated across $R = 3$ repeated evaluation runs, yielding 2,100 end-to-end generation and Ragas evaluation turns (7 configurations $\times$ 100 queries $\times$ 3 runs).

Computational Infrastructure: Neural cross-encoder reranking was executed on an NVIDIA RTX 1060 GPU (4 GB VRAM). Relational curricula are hosted in Neo4j 5 Enterprise with APOC extensions; document parent chunks (2,800 characters, 100 overlap) reside in PostgreSQL, while child chunks (400 characters, 50 overlap) are indexed in Qdrant with `vietnamese-bi-encoder` (768-d). Neural candidate reranking uses `bge-reranker-v2-m3`. Multi-agent coordination was orchestrated via LangGraph under a StateGraph supervisor with Redis 7 caching.

4.2 Evaluation Metrics

Retrieval and Multi-Agent Benchmarks: Retrieval effectiveness (Scenario 1) is measured via standard ranking metrics on top- k returned source docu-

ments: Hit@ k ($k \in \{1, 3\}$), Precision@5, Recall@5, and Mean Reciprocal Rank (MRR@10); domain-stratified analysis reports Hit@1. Multi-agent coordination (Scenario 3) is evaluated across isolated operational decisions: Routing Agent Accuracy and Macro-F1 across the four domains, Intent Classification Accuracy, Tool Selection and Argument Exact Match (EM), End-to-End Pass Rate, Adversarial Tool-Gating Robustness, and Safe Result Behavior under strict operational limits ($\text{max_tool_calls} = 1$, $\text{timeout} = 60$ s).

End-to-End Generative Evaluation: End-to-end synthesis (Scenario 2) is evaluated using the Ragas framework [4] alongside ground-truth regulatory alignment over Top-7 context: Answer Relevancy (AR), Context Recall (CR), Context Precision (CP), and Answer Correctness (AC). In addition, we evaluate Factual Exact Match (Fact EM) for strict numerical tuition fee and credit agreement, Source Recall for gold-source coverage, and Source Average Precision (Source AP) for verified regulatory citations. Evaluating generative QA through LLM-as-a-judge is well-established [21,4]; to mitigate evaluation bias, our framework enforces reference-grounded metrics conditioned on gold documents, greedy decoding ($T = 0.0$) across $R = 3$ repeated runs, and a programmatic Fact EM evaluator for tuition arithmetic.

4.3 Experimental Scenarios and Benchmark Results

To address the three research questions, we evaluate CTU-CHAT across three coordinated experimental scenarios: retrieval component stacking (RQ3), end-to-end QA generation and modular ablation (RQ2), and multi-agent coordination and tool reliability (RQ1).

Scenario 1 — Retrieval Component Stacking (E1–E5): *Objective & RQ Alignment:* Scenario 1 directly addresses **RQ3** by comparing five retrieval configurations at aggregate and complexity-stratified levels on the 100-query Held-Out Test Set. *Configurations:* We compare five cumulative configurations (E1–E5, detailed in Table 2) transitioning from single sparse/dense baselines to hybrid fusion, neural cross-reranking, and our proposed intent-guided evidence stack.

Results: As shown in Table 2, E5 obtains the highest values across the five reported retrieval metrics, reaching 0.7500 Hit@1, 0.9100 Hit@3, 0.2380 Precision@5, 0.8458 Recall@5, and 0.8354 MRR@10. Relative to E4, E5 increases Hit@1 by 7.0 percentage points and MRR@10 by 0.0450; relative to E1, the Hit@1 difference is 22.0 percentage points. These improvements are accompanied by an increase in mean retrieval latency from 15.62 s for E4 to 17.09 s for E5. Because E5 jointly adds intent governance, graph grounding, and parent-child mapping, the E4-to-E5 contrast represents the integrated bundle rather than an isolated routing or graph effect.

The E4-to-E5 Hit@1 differences are largest on entity-comparison and temporal queries (20 percentage points each) and cross-domain queries (15 percentage points). The two configurations tie on multi-hop and adversarial Hit@1, while

Table 2. Scenario 1: Progressive retrieval component stacking on the redesigned Held-Out Test Set ($N = 100$).

Retrieval Configuration	Hit@1	Hit@3	P@5	Recall@5	MRR@10	Latency (ms)
E1: BM25 Lexical Baseline	0.5300	0.7600	0.1980	0.7133	0.6661	9.79
E2: Dense Vector Semantic	0.4200	0.6300	0.1540	0.5542	0.5223	50.80
E3: Hybrid RRF (BM25 + Dense)	0.5000	0.7800	0.2040	0.7308	0.6568	53.10
E4: Hybrid + Neural Reranker	0.6800	0.9000	0.2320	0.8275	0.7904	15,624.33
E5: CTU-Chat Proposed Retrieval	0.7500	0.9100	0.2380	0.8458	0.8354	17,094.94

Table 3. E4-E5 retrieval results by held-out query complexity.

Complexity Tier	N	Hit@1		Recall@5	
		E4	E5	E4	E5
Direct single-hop	40	0.6500	0.6750	0.9000	0.9250
Within-domain multi-hop	20	0.7000	0.7000	0.7167	0.7417
Cross-domain	20	0.7000	0.8500	0.7833	0.7750
Entity comparison	10	0.8000	1.0000	0.9500	0.9500
Temporal	5	0.8000	1.0000	1.0000	1.0000
Adversarial	5	0.4000	0.4000	0.4500	0.5500

E5 obtains higher Recall@5 on both tiers. On cross-domain queries, however, E5 has slightly lower Recall@5 than E4 (0.7750 versus 0.7833), indicating that its improvement in first-rank placement does not extend uniformly to evidence coverage.

Table 4. Scenario 2: End-to-end QA generation and modular ablation on the redesigned Held-Out Test Set ($N = 100$, $R = 3$, 2,100 total evaluation turns, Top-7 context).

Cfg.	CP	CR	AR	AC	Fact EM	Src. R	Src. AP
T1	0.2576	0.5411	0.3961	0.4758	0.5816	0.7308	0.5498
T2	0.1664	0.2682	0.2281	0.3607	0.5443	0.5642	0.4240
T3	0.2110	0.4843	0.3618	0.4631	0.5965	0.7608	0.5469
T4	0.3833	0.6296	0.4454	0.5080	0.6529	0.8408	0.7313
T5	0.2255	0.4617	0.2996	0.4400	0.5857	0.7192	0.6104
T6	0.3761	0.5979	0.4639	0.5108	0.6271	0.8408	0.7058
T7	0.3862	0.6572	0.4235	0.4992	0.6384	0.8208	0.7238

Configurations: T1: BM25; T2: dense; T3: hybrid RRF; T4: full system; T5: without cross-reranker; T6: without Neo4j grounding; T7: without subspace governance. *Metrics:* Fact EM: Factual Exact Match on tuition/credits; CP: Context Precision; CR: Context Recall; AR: Answer Relevancy; AC: Answer Correctness; Src. R: Source Recall; Src. AP: Source Average Precision. Ragas values are means over 300 query-repetition observations per configuration; corresponding standard deviations and bootstrap confidence intervals are retained in the experiment artifacts. All metrics are normalized to $[0, 1]$; higher is better.

Scenario 2 — End-to-End QA Generation and Modular Ablation (T1–T7): *Objective & RQ Alignment:* Scenario 2 directly addresses **RQ2** by

benchmarking the proposed full system (T4) against three single-agent RAG baselines (T1–T3) and three modular ablations (T5–T7) on the Held-Out Test Set (2,100 total evaluation turns; Table 4). *Results & Interpretation:* As reported in Table 4, T4 obtains higher values than the BM25 baseline across all seven reported metrics, including a 7.13-percentage-point difference in Fact EM (65.29% vs. 58.16%). Across all configurations, T4 achieves the highest Fact EM and Source AP and ties T6 for the highest Source Recall. T6 records the highest Answer Relevancy and Answer Correctness, while T7 records the highest Context Precision and Context Recall. The ablation results therefore indicate metric-specific differences rather than uniform dominance by a single configuration.

Scenario 3 — Multi-Agent Routing, Tool Reliability, and Robustness:

Objective & RQ Alignment: Scenario 3 directly addresses **RQ1** by evaluating multi-agent coordination, tool execution reliability, and bounded safety independently from retrieval or generation. *Experimental Setup:* As reported in Table 5, evaluation spans three suites ($N = 550$ runs): (1) **Supervisor Routing (Panel A):** 100 single-domain benchmark queries ($N = 300$ for LLM Supervisor vs. $N = 100$ for Rule-Based Router); (2) **Specialist Tool Reliability (Panel B):** 30 test cases ($N = 90$ runs across 3 repeated runs) covering structured tuition lookup, scholarship calculation, and deterministic tuition reduction arithmetic; and (3) **Adversarial Robustness (Panel C):** 20 failure stress tests ($N = 60$ runs across 3 repeated runs) challenging boundary invariants.

Table 5. Scenario 3: Multi-agent routing, specialist tool execution reliability, and robustness under a maximum of one tool call per decision (max_tool_calls = 1).

Panel A: Supervisor Routing and Specialist Selection ($N = 100$ queries; Rule: 1 rep / LLM: 3 reps, $N = 300$)							
Router Variant	Runs (N)	Agent Acc.	Macro-F1	Intent Acc.	Bounded	Routing Paradigm	
Rule-Based Router	100	85.0%	69.2%	69.0%	100.0%	Heuristic Regex	
LLM Supervisor (CTU-CHAT)	300	97.0%	97.8%	95.0%	100.0%	Structured Pydantic	
Panel B: Specialist Tool Execution Reliability ($N = 90$ runs across 30 test cases, 3 repeated runs)							
Specialist Tool Function	Runs (N)	Selection	Arg. EM	Result Acc.	E2E Pass	Bounded	Paradigm
tuition_lookup	30	100.0%	100.0%	100.0%	100.0%	100.0%	Structured Catalog
scholarship.calculation	30	86.7%	76.7%	86.7%	76.7%	100.0%	LLM Tool Selection
tuition_reduction.calculation	30	100.0%	100.0%	100.0%	100.0%	100.0%	Arithmetic Tool
Panel C: Robustness and Bounded Failure Handling ($N = 60$ runs under malformed/adversarial inputs)							
Adversarial Gate	Runs (N)	Agent Acc.	Tool Gate Safe	Result	E2E Pass	Bounded	Notes
Failure Gate Robustness	60	100.0%	100.0%	98.3%	98.3%	100.0%	–

Note: `tuition_lookup` evaluates deterministic structured catalog queries; scholarship and tuition-reduction cases evaluate LLM-driven parameter extraction and external deterministic tool execution.

4.4 Discussion

Answering RQ1 (Domain-Specialized Multi-Agent Architecture): Centralizing routing under a structured supervisor with asymmetric privileges and bounds (max_tool_calls = 1, timeout = 60s) demonstrated reliable coordination under the evaluated configuration: the supervisor achieves 97.0% agent

accuracy, 97.8% Macro-F1, and 95.0% intent accuracy across 300 runs, outperforming rule-based routing (85.0%, 69.2%). In specialist tool execution (Panel B), `tuition_lookup` and `tuition_reduction_calculation` achieved 100.0% end-to-end pass rates, while `scholarship_calculation` reached 76.7% (86.7% tool selection), with all three functions maintaining 100.0% bounded completion. Furthermore, adversarial testing (Panel C) verified 100.0% tool-gating accuracy, 98.3% safe-result behavior, and 100.0% bounded completion. To assess architectural scalability, a paired stress-test compared CTU-CHAT against a monolithic single-agent baseline: under the baseline registry (11 tools), both architectures exhibited comparable performance (single-agent 93.3% vs. multi-agent 88.3% E2E, difference not statistically significant); however, when 16 synthetic distractor tools were introduced to simulate institutional expansion (27 tools total), the monolithic agent suffered from decision-space pollution, degrading from 98.3% to 91.7% in tool selection. In contrast, domain-partitioned multi-agent maintained 100.0% tool selection (paired $\Delta = +8.3\%$, 95% bootstrap CI $[+1.7, +15.0]$, $n = 60$). The benefit of domain-specialized multi-agent decomposition became more apparent as the tool registry expanded, because partitioned tool access reduced decision-space interference relative to the evaluated monolithic single-agent baseline.

Answering RQ2 (Heterogeneous Knowledge Allocation & Modular Ablations): The modular ablations reveal metric-specific effects. Relative to T5, T4 obtains higher scores on every reported metric, including Answer Relevancy (0.4454 vs. 0.2996), Context Recall (0.6296 vs. 0.4617), Context Precision (0.3833 vs. 0.2255), and Fact EM (65.29% vs. 58.57%), supporting the contribution of cross-reranking within the evaluated configuration. The graph comparison is mixed: relative to T6, T4 has higher Context Recall (0.6296 vs. 0.5979), Source AP (0.7313 vs. 0.7058), and Fact EM (65.29% vs. 62.71%), while T6 has slightly higher Answer Relevancy and Answer Correctness and ties T4 on Source Recall. Similarly, T7 has slightly higher Context Precision and Context Recall than T4, whereas T4 has higher Answer Relevancy, Answer Correctness, Source Recall, Source AP, and Fact EM. These comparisons support component-specific associations within the evaluated setting but do not establish uniform causal effects across query types or metrics.

Answering RQ3 (Representation-Aware Retrieval Effectiveness): On the redesigned benchmark, E3-to-E4 improvements appear in both early-rank ordering and evidence coverage, while E5 provides smaller additional gains across the aggregate metrics. Complexity stratification shows that E5’s largest Hit@1 differences relative to E4 occur on entity-comparison, temporal, and cross-domain queries. Direct and multi-hop queries show smaller or no Hit@1 differences, and the adversarial tier remains difficult for both configurations. Moreover, E5’s cross-domain Recall@5 is slightly below E4 despite its higher Hit@1. Because E5 changes intent governance, graph grounding, and parent-child mapping together, these results compare complete configurations and do not attribute the observed gains causally to any one component.

4.5 Threats to Validity and Limitations

The evaluation is scoped to Can Tho University; multi-institutional validation remains planned. Although the redesigned held-out set explicitly covers multiple complexity tiers, its temporal and adversarial strata contain only five queries each. In addition, the four-domain analysis covers the 80 single-domain queries and excludes the 20 cross-domain queries, so it should not be interpreted as a complete partition of the benchmark. Gemini 2.5 Flash Lite served as both generator and judge, introducing potential evaluation bias despite reference grounding, deterministic decoding, and repeated runs. Curricula graphs rely on offline schemas, and automated judges may penalize valid abstentions. Fault injection across external infrastructure (Neo4j, Qdrant) is reserved for future work.

5 Conclusion and Future Work

This study investigated how centralized supervision and representation-specific evidence paths can support heterogeneous university counseling. The routing and robustness evaluations showed that the supervisor assigned queries to domain specialists with 97.0% agent accuracy while maintaining bounded execution in the evaluated cases. On the redesigned 100-query held-out benchmark, the full retrieval configuration achieved the highest Hit@1 (0.7500), Hit@3 (0.9100), Precision@5 (0.2380), Recall@5 (0.8458), and MRR@10 (0.8354) among the evaluated configurations. Its largest Hit@1 differences relative to E4 occurred on entity-comparison, temporal, and cross-domain queries, although E4 retained slightly higher cross-domain Recall@5. The E4-to-E5 comparison reflects the integrated routing, graph-grounding, and parent-child mapping bundle rather than an isolated component effect. In end-to-end evaluation, the full system achieved the highest Fact EM and Source AP and tied for the highest Source Recall, but did not obtain the highest value on every LLM-judged metric. The ablation results therefore indicate metric-specific contributions rather than uniform superiority of the complete configuration.

Taken together, these findings suggest that no single representation or retrieval strategy dominates all university-counseling tasks. Relational curricula, structured financial information, and narrative regulations impose different evidence and reasoning requirements and are therefore better handled through separate but coordinated processing paths. In practice, the proposed architecture manages these paths within a unified deployment while restricting each specialist to its relevant knowledge sources and tools. The results support the integrated architecture in the evaluated setting, but do not establish that any individual component is universally optimal.

The evaluation remains limited to Can Tho University data, automated judging, offline knowledge schemas, small temporal and adversarial strata, and isolated robustness tests. The single-domain analysis also excludes the benchmark’s 20 cross-domain queries. Future work will expand the held-out benchmark using anonymized real-world student counseling logs, subject to appropriate consent

and privacy safeguards, to improve coverage of naturally occurring intents and conversational variation. Further work will evaluate transferability across institutions, conduct user studies with students and advisors, perform controlled computational-efficiency evaluation and model compression, employ cross-model evaluation to reduce judging bias, and compare CTU-CHAT with broader educational AI systems. It will also expand the production tool registry and evaluate matched single- and multi-agent configurations across increasing tool-set sizes, including production tools rather than synthetic distractors. Within the scope of this study, the central takeaway is not that one retrieval method should replace all others, but that institutional counseling may be better supported when evidence acquisition and execution constraints are aligned with the structure of each domain.

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